

Werner Bisschoff

EMBEDDED SYSTEMS & REAL-TIME SOFTWARE ENGINEER

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Rigorous Systems Engineer with 5+ years of experience engineering real-time, event-driven software architectures (C/C++, FreeRTOS) and secure, deterministic communication protocols. Specializes in designing robust firmware, high-performance IPC mechanisms, and hardware-in-the-loop simulation frameworks. Governed by strict Spec-Driven Development (SDD) methodologies and automated validation pipelines to eliminate runtime faults in production environments.

Experience

Principal Systems Architect

Cape Town

DIVERGENT TABLETOP

Jul 2025

- Integrated a CRDT synchronization layer (`y_ex`) to handle real-time modifications; implemented an actor-based buffering model that reduced state-sync latency to sub-50ms thresholds.
- Designed a dynamic BEAM supervision tree with a `:one_for_one` strategy to isolate concurrent page editing instances, ensuring 99.9% fault isolation and zero system-wide cascading crashes.
- Implemented a threshold-based buffer flush and scheduled heap hibernation (`hibernate_after`), reducing process memory footprints by 60% under active network I/O concurrency.
- Applied Monotropic focus principles and Universal Design to minimize user information processing overhead and ensure highly predictable, deterministic information layout.

Embedded Systems & Integration Engineer

Cape Town

FARO AFRICA

Aug 2024 – Nov 2025

- Designed ISO 14443-4 NFC communication architectures for e-paper display nodes and secure card reader matrices.
- Engineered lock-free asynchronous device management services bridging hardware peripherals with application layers.
- Built Python-based hardware peripheral simulators to catch edge-case device state corruptions prior to flashing.
- Introduced LLM-assisted development workflows to improve debugging speed and code review throughput.

Firmware Engineer (Contract)

Remote

INGENICS DIGITAL GMBH (THROUGH VIVA OUTSOURCING)

Mar 2023 – May 2024

- Architected deterministic FSMs on dual-core ESP32 using FreeRTOS for concurrent sensor arrays.
- Developed Cap'n Proto IPC over serial lines, achieving sub-millisecond serialization latency.

Software Developer

Cape Town

UMAN TECHNOLOGIES

Mar 2021 – Dec 2022

- Implemented RPC services using SOME/IP protocol; reduced bottlenecks with perf.
- Built IPC/RPC layers in C++ and Python using Cap'n Proto; designed node trees for IPC interface.

Junior Lecturer

Potchefstroom

NORTH-WEST UNIVERSITY

Feb 2020 – Dec 2020

- Lectured Python and C++ programming for Introduction to Programming for first year IT students.

Skills

Primary

C++, FreeRTOS, ESP32/ESP-IDF, NFC APDU (ISO 14443-4)

Secondary

NimBLE, UART/SPI/I2C, TinyFrame, Hardware-in-the-Loop (HIL) Testing

Foundational

Assembly, Digital Logic, System Modeling

Other

ERPNext (Integration-focused), Expo Mobile

Education

B.Eng. Computer and Electronic Engineering

Potchefstroom

NORTH-WEST UNIVERSITY

2020

Focus on embedded systems, software engineering, and electronic design · Developed NFC payment protocol emulator (Kotlin/Android), STM32 firmware (C), PID motor controller (Arduino)